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Three bulk density measurement methods provide different results for commercial scale harvests of willow biomass chips

Mark H. Eisenbies[1], Timothy A. Volk[1], Obste Therasme[1], Karl Hallen[1]

[1]- State University of New York, College of Environmental Science and Forestry. 1 Forestry Dr., Syracuse, NY 13210, USA

Abstract

Bulk density is a key attribute of biomass feedstocks that affects both conversion processes and logistics. Reported bulk density for fresh willow chips ranges between 195 and 393 kg m^{-3} and it is unknown if standardized methods using several kg of chips scale effectively to collection vehicles that contain several Mg. The objective of this study was to compare three bulk density methods for 53 commercially harvested willow loads and compare bulk density in three different collection wagons. An ISO standard 'bucket' method was compared to actual and effective bulk density measurements. Mean as-received bulk density using the standard ISO method was 262 kg m^{-3} (σ = 25) and was not correlated with wagon load measurements of actual or effective bulk density. The actual bulk density in a small (10 m^3) collection vehicle was 302 kg m^{-3} (σ = 32), which was significantly greater than the 214 kg m^{-3} (σ = 28) that was measured in two larger (30 m^3) collection vehicles. In addition, actual bulk density in wagons was higher than the effective bulk density (247 kg m^{-3} (σ = 24)) observed for the truck deliveries. Collection wagon geometry and design impacted these results, which may have implications for modeling these systems. The variation in bulk density values is important to understand and represent in modeling of harvesting and logistics systems in order to capture what is actually occurring in commercial scale operations.

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HIGHLIGHTS

- Bulk density (BD) values available in the literature cover a wide range
- BD of shrub willow biomass was determined using three methods
- BD measured in commercial wagons did not match BD values from standardized methods
- Willow BD in collection wagons was influenced by size and geometry of dump bodies
- Effective BD may be an adequate surrogate for actual BD in commercial vehicles

Keywords: Bulk Density; Short-rotation Woody Crops; Shrub Willow; Collection Vehicles; Harvesting Logistics

Abbreviations

SWRC – Short rotation woody crops; BD – Bulk density, BD_{ar} – Bulk density as received, BD_d – Bulk density dry weight basis; BD^{ISO} – Bulk Density using the ISO 17828:2005 standard method; BD^{act} – 'Actual' bulk density determined using methods described in this paper; BD^{eff} – 'Effective' bulk density determined using methods described in this paper; ISO - International Organization for Standardization; MC – gravimetric moisture content on a wet weight basis

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Short rotation willow chip bulk density and differing bulk density estimates based on three determination methods

1. Introduction

Sources of biomass for bioproducts and bioenergy include forests, agricultural crops, various residue and waste streams, and dedicated woody or herbaceous crops [1,2]. Short rotation woody crops (SWRC) are an important source because of their demonstrated high yield potential as well as a capacity to provide additional ecosystem and environmental benefits [3–5]. In SWRC systems, woody plants are managed using a combination of techniques and knowledge from both agriculture and forestry. These systems typically have higher planting densities and more intensive management than traditional forest or forest plantation operations, but do not require annual tillage or frequent applications of fertilizer and herbicides that are typical of annual agricultural crops. Crops in these systems are regenerated by coppice rather than planting and harvested on cutting cycles less than five years [3,6]. Shrub willow is one such crop that has been established as an important SRWC due to its ease of propagation, rapid growth, genetic diversity, and adaptability to a wide range of sites [3,7,8]. Numerous species and cultivars that vary in stem form, productivity, and physical properties have been tested and propagated [7,9].

In the northeastern U.S., shrub willow is generally capable of producing between 8 and 12 Mg ha^{-1} yr^{-1} of dry biomass over three- or four-year rotations [7]. With the variability associated within agricultural fields and management practices, willow crops at the time of harvest can generate anywhere between 20 and 100 Mg ha^{-1} of fresh biomass. There are a

number of approaches and machines for harvesting SRWCs like willow, but the mostly widely used method are single-pass cut-and-chip harvesting operations using a forage harvester paired with collection vehicles [10,11]. The largest of these systems are currently capable of harvesting stems up to 125 mm in diameter and producing green chips of consistent commercial quality in sizes generally less than 63 mm depending on machine settings [12–14]. Collection vehicles can consist of a wide array of truck or tractor/wagon types with many sizes (10 to 115 m^3), heights, and shapes. The choice of collection system is dictated in large part by local availability, field configuration, ground conditions, and the distance between the field and short term storage or end user.

Physical properties are a key aspect of biomass feedstock quality because of their importance not only to pretreatments and conversion processes, but also to the supply chain logistics [15,16]. Among the most basic physical attributes for woody feedstocks characterization, regardless of format, are moisture, particle geometry, and density. Achieving particular chip sizes affects machine performance [13], which is one of several key factors affecting material packing (bulk density). Bulk density (mass per unit volume) influences vehicle selection for collection and transportation of material, and how material is stored and handled [15,17,18]. These system components and stages are critical elements and inputs for life-cycle analyses, and logistic and economic modeling; therefore, it is also crucial to understand their variability as conversion systems may be sensitive to fluctuations in physical properties [18–21].

Bulk density is an ostensibly simple ratio between the weight of biomass and the volume that the material occupies, but it can be challenging to measure consistently. Standardized, non-standard, and undescribed methods are found throughout the literature

where wood chip bulk densities are reported; standardized procedures are designed to produce consistent results for comparison purposes [22]. The ISO 17828:2015 (ISO 17828), EN-15103, and ASAE S269.5 methods are prominent standard methods; each of these methods describe comparable procedures: e.g. filling a container of known volume, shocking it a prescribed number of times, and reporting bulk density either with weight on a dry basis (BD_d; 0% moisture) or 'as received' (BD_{ar}; no moisture correction). The permitted volume of the container is flexible (usually between 0.015 and 0.1 m^3) with some guidelines coupled to particle size, but the height to diameter ratio of the container should fall between 1.25 and 1.5 [22]. Moisture content (MC) and some indication of particle size should also be reported when reporting bulk density.

Reported bulk density values for wood chips from various sources are disparate and vary from material to material and even among apparently similar materials. Willow BD_{ar} values reported in the literature range between 195 and 365 kg m^{-3} for chipped material, and up to 447 kg m^{-3} for finely ground material (Table 1). This is comparatively higher than other SRWCs like poplar or eucalyptus, but less than what is commonly reported for pine or hardwood chips at similar MC, which typically exceed 354 and 454 kg m^{-3} respectively [23]. One issue with reported bulk densities is reconciling the array of individually reported values with the MC and particle sizes and methodology used. Unfortunately, whether BD_d or BD_{ar} are being reported and what method was used is not always explicit, and given the wide range of values reported, it can in some cases be difficult to infer without some ambiguity which value is being presented [9,15,17,24].

105 [INSERT TABLE 1]

106

107 While bulk density values generated using standardized methods may be useful in
108 ascertaining qualitative differences between materials with respect to harvesting systems or
109 processing, it is less apparent how well bulk density measured in relatively small containers
110 scales to large volumes or containers. The work by Böhm and Hartman [22] specifically implies
111 that bulk density for materials like wood chips may be sensitive to the measurement
112 conditions. Although container configuration appears to make a difference [22], it is also
113 uncommon for biomass bulk density to be determined at relevant scales like full-size vehicles
114 [25]. Generally an available value for bulk density from the literature is scaled to a vehicle or
115 some other part of the supply chain because that is the only value that is available [15,17,18].
116 Shrub willow is an important feedstock and more information is needed about its
117 physical properties in the context of the supply chain, particularly transport, in order to model
118 and optimize these systems. Ignoring the highest and lowest reported values for willow in the
119 literature places BD_{ar} values between 250 and 390 kg m^{-3}. Preliminary and informal work
120 previously conducted by the authors has suggested that two standard deviations of BD_{ar}
121 determined using standard methods or estimates based on vehicle volumes spans a range
122 between 241 and 343 kg m^{-3} (mean 289 kg m^{-3}). The wide range of BD values found in the
123 literature, the varying size and array of collection vehicles used in harvesting willow, and the
124 wide range of previously observed values by this group suggest a need to better understand the
125 relationship between bulk density, measurement methodology, collection vehicle type, and
126 crop characteristics. The objective of this study is to determine bulk density of willow biomass

chips harvested with a single pass cut-and-chip system using three methods and to evaluate the influence of collection vehicle type and standing biomass on bulk density in the context of harvesting and transport.

2. Materials and methods

2.1. Study Plan

The study plan for this work entailed determining bulk density of individual loads of willow chips from collection vehicles used on a commercial harvest using three methods of bulk density determination: an ISO method (BD^{ISO}), a direct measure of actual volume and weight of chips (actual bulk density, BD^{act}) and a measure that used estimated volume based on the empty volume of the collection wagon and a measured weight (effective bulk density, BD^{eff}). Regression was used to determine relationships among bulk density measurements using the different methods. Cofactors such as standing biomass of the harvested crop, collection vehicle volume and suspension type were also evaluated.

2.2. Site and Harvests

Harvesting occurred on a cluster of seven short rotation willow fields (designated A through G) as part of a larger deployment of crops located near Cape Vincent, NY (44.126N, 76.318W). The mixed-cultivar crop was established in the spring of 2013 [26], coppiced in the spring of 2014, and had completed three growing seasons through the summer of 2017. Field sizes ranged between 1.6 and 10.4 ha for a total of 32.1 ha; row lengths varied between 200 and 450 m. A wide expression of heights and standing biomass were available for harvest due to differences in crop management, cultivars planted, and varying site quality. In a quick pre-harvest crop survey heights measured across all stands ranged from 4 m to just over 8 m tall

which suggested to operators the yields would likely range between 15 and 100 Mg ha^{-1} across the entire site. This information was used to plan which loads would be utilized for this bulk density study in order to stratify loads/samples across the full range of standing biomass available. For this study, 53 wagon loads generally stratified by standing biomass (based on height) were collected over several days with material coming from loads well distributed across all fields.

Willow for the bulk density analyses was harvested leaf-on over a four day period, September 5-8, 2017. The operation was conducted as a single-pass, cut-and-chip process using a New Holland FR-9080 forage harvester equipped with a New Holland 130FB coppice header. The length of cut selected by the operator was the largest setting ("33-mm") in order to maximize fuel economy and harvesting rate [13]. Material was cut, chipped and blown into one of three tractor-drawn, high-lift dump wagons: “small” (≈10 m^3 Badger 120 with fixed, two-axle suspension (Miller St. Nazianz Inc., Nazianz, WI)), “walking” (≈30 m^3 Broussard Model 4408 high lift sugar cane wagon with ‘walking’, dual-axle suspension (Broussard Cane Equipment, St Martinville, LA)), or “fixed” (≈30 m^3 Broussard high lift sugar cane wagon with fixed, two-axle suspension, model unspecified (Broussard Cane Equipment, St Martinville, LA)). Harvest and collection equipment operations were subsequently monitored using Trimble GPS devices (Geo 7 and Juno 3B series), using the methodology described in Eisenbies et al [12]. GPS data was used to calculate actual biomass yields and harvesting rates.

2.3. Bulk Density Determination

Bulk density for 53 wagon loads was determined using one to three methods for calculating bulk density on an as-received and dry weight basis: (1) standard bulk density using

ISO 17828 method (BD_{ar}^{ISO}, BD_{d}^{ISO}), (2) actual bulk density by measuring the actual volume of chips carried by a collection vehicle and the weight of the chips in the wagon (BD_{ar}^{act}, BD_{d}^{act}), and (3) effective bulk density by estimating the volume of chips based on the fixed volume of the collection vehicle dump body and the weight of the chips in the wagon (BD_{ar}^{eff}, BD_{d}^{eff}) [27]. The BD^{ISO} values were determined on 48 loads (a subset of the 53 total loads) using a 0.0189 m^3 cylinder with a 1.29 height to diameter ratio. Chipped material for the BD^{ISO} measurement was taken from six stratified locations (front to back) of the exposed chips after each wagon load deposited its contents on the short-term storage pile. MC was determined gravimetrically using a 1-2 kg subsample of the material utilized for the ISO method bulk density measurements [28]. Wet weight was measured within 30 minutes of collecting the material, delivered to a lab the same day, and dried at 65°C to a constant weight. Given the focus harvesting and transport, analyses will be presented in this paper are as received using wet weights to determine bulk density (BD_{ar}) unless otherwise noted.

The actual bulk density (BD_{ar}^{act}) for 38 loads (a subset of the 48 BD^{ISO} loads) was determined by weighing and measuring the volume of chips being transported by a collection vehicle. The weight of biomass was determined using a four-point platform truck scale (Cardinal Scale Manufacturing Company, Webb City, MO) weighing each axle of the collection vehicle empty when they enter the field to obtain the tare, and again as they exited the field to obtain the loaded weight. The volume of collected biomass was determined using a TruPulse 360R laser survey instrument (Laser Technology Inc., Centennial, CO) that took multiple readings (n>25) to obtain three 3D surfaces: the empty dump body interior (one time), the top plane (the top opening of the dump body), and the 3D mound profile of the loaded chips

(including points below the top plane at the edges). The three surface measurements were taken from a telescoping construction boom lift that raised the laser and operator to a height of approximately 10 m above the ground and permitted the necessary perspective to observe the interior portions and surfaces of the dump body. The difference between the top plane and mound profile was subsequently used to calculate the volume of the voids (the unfilled portions of the dump body down to the chips) and the volume of the mounded portion of chips. A simple summation of the void, overfill, and dump body volumes was used to calculate the actual biomass volume. The laser methodology was verified by comparing the empty volume determined using the laser against a physical measurement of the dump body dimensions. Differences were less than 5% for all three of the wagons used.

The effective bulk density ($\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$) is a simple ratio of the weight of fresh biomass to the fixed volume of the measured dump body when it is empty; it was determined for all 53 loads. The top edge of the dump wagon is used to calculate the volume and the voids that may be created when the wagon is filled or as it travels across the field or mounds created from overfilling are not accounted for. Regardless, $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$ is the simplest of the three values to obtain and it was included because it may serve as an adequate surrogate for BD^{act} or BD^{ISO} in terms of biomass logistics; granting that BD^{eff} is merely a scaled transformation of load weight expressed as a density.

2.4. Particle Size Analysis

Moisture content samples from each field were composited to determine the particle size distribution of chips. Samples utilized for MC were combined and reduced to a 0.75 to 2.0 kg quantity using a Gilson SP1 universal sample splitter (Gilson Company, Lewis Center, Ohio) to

ensure particle size distributions were preserved. Particle size separation was accomplished using a modified Williams wood chip classifier following a procedures outlined in Eisenbies et al, 2014 [29]. The modification utilizes standard sieves with circular holes (3.15, 15.9, 25.4, 31.8 mm) coupled with 12-mm thick plywood plates with matching hole-patterns for each screen size. This procedure is meant to prevent spearing; a problem with SRWC due to the small stem diameters that result in cylinders or split cylinders when harvested. Sieving time for each sample was 15 seconds. After sieving, trays were hand-sorted to find chips that speared in spite of the anti-spearing plates. Chips were returned to their appropriate tray size based on the longest dimension of the chip before weighing. Also, chips between 31.8 and 63 mm for samples were sorted by hand using a gauge and the maximum chip length and maximum cross section for each sample were determined by hand using a ruler to establish compliance with the ISO 17225-4 P45S standard for graded wood chips.

2.5. Statistical Analyses

Regression was used to evaluate the relationships between the three BD methods used, or between BD and crop conditions, moisture content, vehicle size, or suspension. Statistical comparisons of the bulk density methods and covariates (i.e. standing biomass, MC, height, wagon type) were conducted as simple and multiple linear regression models in SAS 9.4 (SAS Institute, Cary, NC) using the REG procedure. Each individually collected and observed load of chips was considered independent and used as the experimental replicate. Four full models were evaluated to examine $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ or $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ in the context of method, wagon type, and crop conditions:

Model 1:

$$\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}} = \beta_0 + \beta_1 BM + \beta_2 Ht + \beta_3 MC + \beta_4 BM^2 + \beta_5 Ht^2 + \beta_6 MC^2$$

Model 2:

$$\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}} = \beta_0 + \beta_1 \mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}} + \beta_2 \mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}\,2} + \beta_3 Size + \beta_4 \mathrm{Size} \cdot \mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}} + \beta_5 \mathrm{Size} \cdot \mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}\,2}$$

Model 3:

$$\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}} = \beta_0 + \beta_1 \mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}} + \beta_2 \mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}\,2} + \beta_3 Size + \beta_4 \mathrm{Size} \cdot \mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}} + \beta_5 \mathrm{Size} \cdot \mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}\,2}$$

Model 4:

$$\begin{aligned}\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}} = {} & \beta_0 + \beta_1 BM + \beta_2 Ht + \beta_3 MC + \beta_4 BM^2 + \beta_5 Ht^2 + \beta_6 MC^2 + \beta_7 Wt + \beta_8 Wt^2 \\ & + \beta_9 Walk + \beta_{10} Walk \cdot BM + \beta_{11} Walk \cdot Ht + \beta_{12} Walk \cdot MC \\ & + \beta_{13} Walk \cdot BM^2 + \beta_{14} Walk \cdot Ht^2 + \beta_{15} Walk \cdot MC^2 + \beta_{16} Walk \cdot Wt \\ & + \beta_{17} Walk \cdot Wt^2\end{aligned}$$

Where

$\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$: Bulk density as determined using ISO 17828 standard (kg m^{-3})

$\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$: Actual bulk density as determined by methods described in this paper (kg m^{-3})

$\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$: Effective bulk density as determined by methods described in this paper (kg m^{-3})

BM: Standing biomass delivered for each load as determined using crop area from the GPS analysis and weighed mass of the chips (Mg ha^{-1})

MC: Gravimetric Moisture Content (%)

Ht: Average stand height for field of origin as determined by initial survey (m)

Wt: Weight of loaded chips (kg)

Size: Dummy variable designating small or large dump wagon (0,1)

Walk: Dummy variable designating fixed or walking suspension (0,1)

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These regression models are not intended for use as predictive functions, rather, to determine whether the three methods of BD are interchangeable and whether the class and continuous factors in this experiment influence the bulk density of harvested loads. The progression of model development address the applicability of each method of BD determination in terms of the stated objectives, address their interchangeability, and address the stated objectives specifically. First, model 1 evaluates whether there is a relationship between $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ and crop attributes; vehicle size and suspension was not considered because material was collected after it had been dumped on the storage pile. Models 2 and 3 were used to evaluate whether there relationships between $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ and $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ or $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ and $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$; wagon size was added to these models since it became apparent in the initial assessment of $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ that the bulk densities associated with wagon size were very different. Model 4 was a comprehensive evaluation of $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ relative to crop and vehicle attributes.

Model selection and development for multiple linear regression models was conducted using backwards selection and the Mallows Cp statistic; the alpha level was 0.05 [30,31]. Potential outliers were assessed, but there were no justifiable reasons for removing observations from the reported data and their removal did not fundamentally alter conclusions relative to the objectives. During model 4 development, observations from the small and large dump wagons were modeled separately because their inclusion imposed extreme leverage and interference with regards to the suspension and crop attributes. In other words some preliminary modeling did include a “Size” and its interactions, but that model was over-specified and so different it prevented a coherent comparison of the two cane wagons. Checks

for collinearity between dependent variables were evaluated using the "variance inflation factor" statistic [31]. Normality assumptions were tested using the UNIVARIATE procedure, summary statistics were calculated using the MEANS procedure, and any means comparisons were determined using the GLIMMIX procedure in SAS 9.4.

3. Results and Discussion

3.1. ISO Bulk Density

Bulk density calculations are typically adjusted for MC as a means of comparing raw materials from different sources. Because of the relatively low variation in MC of willow at the time of harvest and consistency in particle sizes produced from the harvester, the BD_{ar} and BD_{d} are well correlated (all P values <0.0001; all R^2 values >0.8746) for all three methods of bulk density determination (Figure 1). Thus, relationships observed or conclusions drawn from either an as-received or dry context will be similar. BD_{ar} is a more relevant number than BD_{d} in terms of the logistical aspects of moving this material through supply chain.

[INSERT FIGURE 1]

Values for BD_{ar}^{ISO} had a mean of 262 kg m^{-3} (n=48, σ=25) and ranged from 215 to 310 kg m^{-3}; generally they were lower than reported in the literature. This may be due in part to the portions of the field that were undersized based on the estimated standing biomass. An acceptable annual yield for large scale production of willow is generally thought to be above dry 8.5 Mg ha^{-1} yr^{-1} [3,7], which is approximately 50 Mg ha^{-1} of fresh biomass at the time of harvest. For loads that were obtained from areas above a 50 Mg ha^{-1} threshold, the mean BD_{ar}^{ISO} was

283 kg m^{-3} (n=14, σ=13) and ranged from 257 to 301 kg m^{-3}. In contrast, for loads that were obtained from areas below a 50 Mg ha^{-1} threshold, the mean BD_{ar}^{ISO} was significantly lower at 253 kg m^{-3} (n=32, σ=23, P<0.0001).

BD_{ar}^{ISO} was highly correlated to standing biomass (P=0.0071, R^2=0.7640). The final version of model 1 also established stand height (P=0.0012) and MC (P=0.0061) as significant regression components; other components of model 1 were not significant (Figure 2). Taller stems coupled with high biomass tended to yield higher bulk densities using the standardized ISO method. Samples with higher MC also, unsurprisingly, resulted in higher bulk density. MC had a mean of 52.9% and ranged from 42.7 to 60.2%, which is higher than previously reported for willow but this harvest occurred when leaves were still on the plants. BD_{ar}^{ISO} increased 12.48 kg m^{-3} for each meter of stand height and 2.19 kg m^{-3} for each percent increase in MC. Monerdero et al. [24] observed that coarser chips for typical short rotation woody crops, tend to have higher bulk densities. However, the fresh leaves and small twiggy pieces that are found in these loads may enhance bridging that imparts a certain springiness or resilience to the tamping that occurs during bulk density determination [32]. A greater quantity of course material was observed in material that was obtained from the fields with larger trees (Table 2).

[INSERT TABLE 2]

[INSERT FIGURE 2]

For this harvester and the settings used, the most applicable target particle size standard for wood chips is the ISO P45S [29]. In this and previous work, this machine easily

complies with the fine limits (<10% by weight), and the oversize limits by not exceeding a 200 mm maximum dimension or 6 cm^2 maximum cross section permitted (Table 2). The distribution of the course fraction size classes appear to be dependent on the standing biomass; larger trees result in a larger proportion of large chip sizes. Due to the labor intensiveness obtaining accurate particle size distributions for this type of material, the data is not available for more rigorous examination of these possibilities. The leaf-on harvested material ultimately complies with the ISO P45S standard and various components of the course fraction may influence bulk density in a variety of ways outside of their physical dimensions.

3.2. Actual Bulk Density

Values for $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ had a mean of 240 kg m^{-3} (n=39, σ=50) and ranged from 162 to 362 kg m^{-3}; generally they were much lower than reported in the literature. Considering the same 50 Mg ha^{-1} level of desirable production for willow biomass as before, the mean fell to 228 kg m^{-3} (n=10, σ=38) and the range closed to 165 to 304 kg m^{-3}, and in contrast, the mean was 243 kg m^{-3} (n=28, σ=54) for stands under 50 Mg ha^{-1}. $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ and $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ are not well correlated (R^2=0.0051) (Figure 3a). The only significant component of model 2 was wagon size (P<0.0001).

[INSERT FIGURE 3]

The mean $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ was 89 kg m^{-3} higher (P<0.0001) for the small dump wagon compared to the two cane wagons (Table 3). This high bulk density observed for the small dump wagon may be due to a variety of reasons. (1) Both fixed and walking axle cane wagons are relatively tall high compared to the harvester's boom; chips are blown against the interior of the far wall

of the wagon and then they slide down the wall and settle into the wagon. The small Badger wagon is shorter so the chips from the harvester are blown down into the bottom of the wagon and on top of other chips as they accumulate. (2) The side of the cane wagons that chips were blown against was a rough surface made of expanded metal sheet with approximately 16 x 38 mm holes. As chips were blown against the side, an unknown proportion of fines may have passed through the grating. In addition, the grating is rough and may reduce the amount of settling, resulting in reduced BD. The small dump wagon had solid sheet metal on all four sides so all the chips were contained in the wagon and there was less friction between the material and sides. (3) Along with the expanded sheet sides providing friction the geometry of the fixed and walking axle cane wagons interiors may have prevented settling and compaction. The cane wagons had inclusions in the bottom of the wagons to make room for the axels and this framework may have contributed to bridging in the bottom. In contrast, the small dump wagon had a smooth, V-shaped monolithic interior which may increase compaction. (4) All dump wagons had different suspension systems. The fixed suspensions of the small dump wagon and one of the Broussard wagon likely experienced a higher degree of shock loading during traverses compared to the walking axle cane wagon. (5) Although we have no means to illustrate this difference, the operator of the tractor that pulled the small dump wagon tended to be more aggressive over all types of terrain. The cane wagons had higher centers of gravity and greater inertia, which required operators to traverse rough terrain more carefully. As a result the loads in the small dump wagon may have experienced more substantial shock loads due to operator behavior.

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362 [INSERT TABLE 3]

363

364 The relationships between collection wagon geometry, shock loading, material bulk
365 density, material compressibility, particle size and shape, and MC are complicated [33,34]. The
366 distinct and significant differences found between the smaller and larger wagons was not
367 expected and determining the specific reasons, and their importance, leading to observed
368 differences is generally beyond the scope of this work and would require additional studies.
369 However, Böhm and Hartman [22] demonstrated that container size (0.015-0.1 m^3 range) and
370 configuration as well as shock loading are important considerations when determining bulk
371 density. Their results suggest that decreasing container size decreases bulk density estimates
372 by around 10%. In the case of this study, the inverse appeared true; the smaller container had
373 higher bulk density by approximately 40%. It is possible, and maybe not surprising, that
374 transferring bulk density results based on lab containers, such as the implication from Böhm
375 and Hartman [22] do not translate well to much larger collection containers and indeed this
376 was apparent in this study. Small containers with small sized and uniform sizes that are used
377 with a standard amount of shocking do not effectively represent the BD of operational
378 containers where factors such as geometry, shocking, and roughness vary.

379 For instance, there were differences in the relationship between load weight and the
380 $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ for the different collection wagons. For the fixed and walking axle cane wagons the$\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$
381 increased 21.3 kg m^{-3} for every Mg increase in the load ($P<0.0001$). One inference is that the
382 chip bulk density increases under their own weight (Model 4; Figure 4). However, in the case of
383 the small dump wagon there was no relationship between increased relative to load weight

(P=0.5255), and no other factors were significant in model 4. The range of load weights for the small dump wagon was small ($\hat{c}_v$ = 9.5) compared to the fixed and walking axle cane wagons ($\hat{c}_v$.= 18.0). As stated previously, this may be due to the way the wagons were driven, indicative of chips being blown into the dump body from different angles This fact may explain why a weaker relationship exists between $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ and standing biomass (P=0.0415), yet was the driving factor for $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ (Model 1). The springiness of the leaves and twigs in a 0.0189 m^3 cylinder potentially affecting the $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ result is probably not a factor under the weight of several tons of biomass in a wagon that is shifting as it moves across the field.

[INSERT FIGURE 4]

Once again, Böhm and Hartman [22] also showed that shock loading in their experiments resulted in 10-12% higher bulk densities. Although the suspension type did not result in significantly different $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ values or interactions (P=0.1219), there was a numerical 5% higher $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ value for the cane wagon with the fixed suspension than the walking suspension. This seems to corroborate Böhm and Hartman's [22] result since a fixed suspension probably has greater shock loading, but not significant enough to distinguish or recommend one wagon based on suspension type alone.

3.3. Effective Bulk Density

Values for $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$ had a mean of 251 kg m^{-3} (n=53, σ=57) and ranged from 140 to 393 kg m^{-3}; as with $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$, they were much lower than reported in the literature. Considering the same 50 Mg ha^{-1} level of desirable production for willow biomass as before, the mean and range

above this cut off, 235 kg m^{-3} (n=14, σ=65) and 144 to 354 kg m^{-3}were simialr to values for where standing biomass was less than 50 Mg ha^{-1}, 256 kg m^{-3} (n=39, σ=54). BD_{ar}^{eff} and BD_{ar}^{act} are well correlated across all wagon types (R^2=0.8505) (Figure 3b). BD_{ar}^{eff} for the small dump wagon had a weak correlation (R^2=0.1528) with BD_{ar}^{act} while relationship for the cane wagons was much stronger (R^2=0.7523). These patterns suggest that BD_{ar}^{eff} is not always an acceptable surrogate on an individual load basis, but are a reasonable estimate over many loads (Table 3). Regardless, BD_{ar}^{eff} may better represent loads coming off the field than BD_{ar}^{ISO} with the benefit of being relatively simple to obtain.

3.4. Sources of error

Distributions for three methods for assessing bulk density spanned wide ranges of values. BD_{ar}^{act} and BD_{ar}^{eff} are values for bulk density using techniques that have not been standardized. The means and ranges for both were sufficiently different on a load basis from BD_{ar}^{ISO} and lower than what was reported in the literature to warrant some discussion of the potential sources of error. These results alone demonstrate the potential variability associated with bulk density and the importance of not relying on a single mean where that variability could have implications on harvesting and logistics planning and costs.

An obvious potential source of error could be associated with the scale used to weigh the loads. The portable scales were located on generally firm, level ground, but not a cement pad. Care was taken to ensure the wheels of the wagons were centered on the scale. A check weight for a full load was compared against certified scales located at a nearby rock quarry. The difference between these measurements when the load was centered was below 5%, which means our values for BD_{ar}^{act} and BD_{ar}^{eff} would vary by approximately 5% if there were

errors associated with the field scale. However, this type of error does not sufficiently explain the magnitude of differences observed between $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ and $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$. Weight discrepancies would be irrelevant to comparisons of $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ and $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$ since that would only entail a scaling error. In addition, mean $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ for the various wagon types bracketed the values for $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$. A weighing error would not adequately account for the variation in the ranges that were observed or necessarily change many of the conclusions presented in this paper.

Another weighing error would include issues associated with accommodating the varied suspensions and wheel locations of each collection vehicle type. We evaluated this post hoc by conducting a paired t-test of consecutive loads of biomass into a cane wagon, and then into the small dump wagon. The assumption is that consecutive loads from the same part of the field would have similar standing biomass. Given that each wagon type traverses a different distance for the biomass collected, if there was an issue with weighing each collection vehicle type the estimated standing biomass would differ significantly depending on the collection vehicle. There was no significant difference in standing biomass estimates (P=0.1151), which suggests that the scales performed similarly for each wagon type on consecutive loads. Additionally, the mean and range of bulk densities for the small dump wagon were like those for a similarly constructed, 15 m^3 Richardton 700 (Sunflower Manufacturing, Beloit, KS) for previous work on another harvest site (unpublished $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$ data).

A third source of potential error could be associated with the volume determination of each collection vehicle type. The laser device used for this study had a precision of 30 cm. The precision issue was accounted for to a degree by shooting 25 or more position points for laser measurements. In addition the dump body dimensions were hand measured using ruled

devices. Based on our results there may have been up to a 5% under-estimate of dump body volumes. This translates to a 15 kg m^{-3} shift in bulk density estimates. Once again, this is predominantly a scaling error that would not adequately explain the differences observed between $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$, $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$, and $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$ for the two types of dump wagons. It would also be irrelevant to relationships concerning to $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{act}}$ and $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$.

Given that a relationship between $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ and standing biomass was observed, which may have been due in part to bridging in samples with more leaves and twigs, a final hypothesized source of error was that material collected from the field edges differed from material collected from the field interiors. Willow at fields edges tends to have higher amounts of twiggy and leaf material due to their growth behavior with additional sunlight. There were no significant difference detected in the bulk density values due to an edge effect among fields or wagons for any models tested.

3.5. Implications

It is apparent from the results in this study that bulk density values reported in the literature or determined using the standardized methods may not represent the willow chips collected and transported in association with large-scale harvesting. This may be due to the specific issues with size and geometry of chips sourced from multi-stem small diameter plants, and in this case, the crop was harvested leaf-on. In any case, scaling of bulk density from small containers to large collection vehicles is not a given for shrub willow harvested with leaves on.

Additionally, if the material compresses or dries en route, effectively changing the bulk density, additional material cannot be added or removed; thus, only the weight of chips physically on board is relevant and changes in BD are not necessarily important from a

transport standpoint. Mounded material can rise substantially over the tops of the dump bodies, so the reported dump body volume coupled actual bulk density may not reflect the real capacity of a vehicle for modeling purposes. For instance, the cane wagons tended to have an additional 4-6% of volume due to overfill, as opposed to about 1% overfill for the small dump wagon (Table 3).

On public roads there may be limits to the weight of material that can be legally transported. For example, in the U.S. the legal limit for a common 5-axle tractor trailer is 36.2 Mg, with 5.44 Mg allowed on the front axle and the remaining weight distributed to the rest of the truck [35]. Bed sizes for most tractor trailers typically range between 90 and 115 m^3, and empty weights for trucks may range between 15.8 and 20.1 Mg. The rules and limits otherwise vary by state, axle type, axle spacing, and permits or exemptions obtained. The truck that was used for to deliver material to the end user in this study had a mean BD_{ar}^{eff} value of 247 kg m^{-3} (n=63, σ=24) and ranged from 199 to 307 kg m^{-3}; similar to what was observed for the cane wagons and much lower than reported in the literature (Tables 1 and 3) and 15 kg m^{-3} less than the values obtained using the ISO method in this study.

In practical terms, a 90 m^3 trailer fully loaded with the observed ranges of BD_{ar}^{act} and BD_{ar}^{eff} would have an average payload weight of 22.2 Mg and range between 10.7 and 27.6 Mg; the truck used in this study the range was between 17.9 and 27.6 Mg. Similarly, a 115 m^3 trailer fully loaded with this range of bulk densities would have an average payload weight of 28.4 Mg and range between 22.9 and 35.3 Mg. Thus, the empty weight of a truck and size of the bed are important considerations for transport of this material, but also changes that may occur to the material, particularly moisture content, during on-truck storage since trucks may exceed

494 the unpermitted weight limit. However, it is also true that average estimates for willow bulk
495 density available in the literature are higher than those presented in this study. As a result,
496 analyses that rely on previously published values to establish payload limits may be under
497 predicting the amount of biomass transported per truckload depending on the assumptions
498 made.

499 The amount of variation in the BD values measured in this study is large and was
500 impacted by a number of factors that have been noted. It is important to be aware of and
501 include this variation, and not just mean values, in models that are used for large scale
502 operations like 1,000 to 2,000 Mg day^{-1} facilities that are often considered for expansion
503 [36,37]. A change in the bulk density of this material could have important impacts on the
504 amount of equipment needed to collect this material during harvesting operations and move it
505 to depots or end users. Harvesting and transportation are the two largest cost factors for
506 SRWC, so these steps in the logistics supply change need to be modeled as accurately as
507 possible.

508 **4. Conclusions**

509 Bulk density is an important value for decision making that has implications throughout
510 the supply chain so it is important to have values based on measurements from equipment that
511 would be used for commercial scale operations. Based on this study single values for bulk
512 density obtained from the literature should be used with some caution and understanding that
513 for a material like SRWC may be quite variable depending on a number of factors such as
514 collection equipment, crop characteristics, and field conditions.

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The reader may ask which value, range, or method are recommended based on these results; it depends. If the application is to contrast willow biomass with other biomass we would recommend using the standardized methods such as the ISO bucket values presented here. Standardized methods are widely use and they probably highlight or encapsulate certain qualitative difference between feedstocks that may be important to end users or conversion processes. Mean $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ determined using standardized method was 262 kg m^{-3} (σ = 25), which were lower than elsewhere in the literature, probably because this material was obtained from a leaf-on harvest conditions and some of the stands had standing biomass under 50 Mg ha^{-1}. As noted above the use of the use of the range of values measured and not just a single mean value is an important consideration.

If collecting bulk density values for an ongoing commercial harvesting operation, we would tend to endorse the use of $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$ (the simple ratio of load weight to the fixed volume of the dump body) since it is easy to determine and comparable to $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$. Mean $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{eff}}$ was 250 kg m^{-3} over 57 loads and 235 kg m^{-3} where standing biomass was above the commercially desirable 50 Mg ha^{-1} threshold.

If the application is to model harvesting, logistics and delivery systems for SRWC, we would suggest using the range of values for actual or effective bulk density since those are comparable and the most relevant to the movement of materials. However results suggest that there is a great deal of potential variability associated with standing biomass, and vehicle configurations that should be considered and accounted for in modeling efforts.

These results do suggest that BD obtained using standardized methods may not be representative of materials in a harvesting and logistics setting when loads are in the range of a

3 - 10 Mg and carried by dump bodies in the $10m^3$ to $30m^3$ range. In essence, estimating bulk density using a small volume container, as is done in the standard method used in this study, does not accurately reflect the bulk density in a large collection vehicle, especially when the chipped material is blown into the wagon. It is also apparent that material's bridging behavior, which may influence $\mathrm{BD}_{\mathrm{ar}}^{\mathrm{ISO}}$ values, can compress under its own weight on larger dump bodies. Additionally, design elements of collection wagons such as geometry, wall construction, and suspension may influence bulk density, and that may change when the material is dumped on short term storage or loaded onto trucks.

Finally, there is variation in each of the BD values measured in this study and this needs to be considered in the analysis of these systems. The high end of the observed effective bulk density values indicate that tractor trailers could exceed load limits for public roads if their entire volume was filled. Logistic models that include load regulations as part of their methodology, and utilize existing published bulk density values for willow as determined by standardized methods could underestimate that can be transported.

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References

[1] U.S. Department of Energy, 2016 Billion-Ton Report: Advancing Domestic Resources for a Thriving Bioeconomy, Volume 1: Economic Availability of Feedstocks, ORNL/TM-2016/160 (Internet), Oak Ridge National Laboratory, Oak Ridge, TN, 2016 (cited 2017 Jun 20).

Available from: https://energy.gov/sites/prod/files/2016/12/f34/2016_billion_ton_report_12.2.16_0.pdf.

[2] N. El Bassam, Handboook for bioenergy crops, Earthscan, London, 2010.

[3] T.A. Volk, J.P. Heavey, M.H. Eisenbies, Advances in shrub-willow crops for bioenergy, renewable products, and environmental benefits, Food Energy Secur. 5 (2016) 97–106. doi:10.1002/fes3.82.

[4] R.S. Zalesny, J.A. Stanturf, E.S. Gardiner, J.H. Perdue, T.M. Young, D.R. Coyle, W.L. Headlee, G.S. Bañuelos, A. Hass, Ecosystem Services of Woody Crop Production Systems, BioEnergy Res. 9 (2016) 465–491. doi:10.1007/s12155-016-9737-z.

[5] D.L. Rockwood, C.V. Naidu, D.R. Carter, M. Rahmani, T.A. Spriggs, C. Lin, G.R. Alker, J.G. Isebrands, S.A. Segrest, Short-rotation woody crops and phytoremediation: Opportunities for agroforestry?, in: New Vistas Agrofor., Springer, 2004: pp. 51–63.

[6] D. Dickmann, Silviculture and biology of short-rotation woody crops in temperate regions: Then and now, Biomass Bioenergy. 30 (2006) 696–705. doi:10.1016/j.biombioe.2005.02.008.

[7] N.J. Sleight, T.A. Volk, G.A. Johnson, M.H. Eisenbies, S. Shi, E.S. Fabio, P.S. Pooler, Change in Yield Between First and Second Rotations in Willow (Salix spp.) Biomass Crops is Strongly Related to the Level of First Rotation Yield, BioEnergy Res. 9 (2016) 270–287. doi:10.1007/s12155-015-9684-0.

[8] T.A. Volk, L.P. Abrahamson, K.D. Cameron, P. Castellano, T. Corbin, E. Fabio, G. Johnson, Y. Kuzovikina-Eischen, M. Labrecque, R. Miller, D. Sidders, L.B. Smart, K. Staver, G.R. Stanosz, K. Van Rees, Yields of biomass crops across a range of sites in North America, Asspects Appl. Biol. 112 (2011) 67–74.

[9] B. Caslin, J. Finnan, A.R. McCracken, Willow varietal identification guide, Agrifood Biosciences Institute, Belfast, Ireland, 2012.

[10] S.P.P. Vanbeveren, R. Spinelli, M. Eisenbies, J. Schweier, B. Mola-Yudego, N. Magagnotti, M. Acuna, I. Dimitriou, R. Ceulemans, Mechanised harvesting of short-rotation coppices, Renew. Sustain. Energy Rev. 76 (2017) 90–104. doi:10.1016/j.rser.2017.02.059.

[11] R. Spinelli, E. Cacot, M. Mihelic, L. Nestorovski, P. Mederski, E. Tolosana, Techniques and productivity of coppice harvesting operations in Europe: a meta-analysis of available data, Ann. For. Sci. 73 (2016) 1125–1139. doi:10.1007/s13595-016-0578-x.

[12] M.H. Eisenbies, T.A. Volk, J. Posselius, C. Foster, S. Shi, S. Karapetyan, Evaluation of a Single-Pass, Cut and Chip Harvest System on Commercial-Scale, Short-Rotation Shrub Willow Biomass Crops, BioEnergy Res. (2014). doi:10.1007/s12155-014-9482-0.

[13] S.P.S. Guerra, G. Oguri, R. Spinelli, Harvesting eucalyptus energy plantations in Brazil with a modified New Holland forage harvester, Biomass Bioenergy. 86 (2016) 21–27. doi:10.1016/j.biombioe.2016.01.003.

[14] J. Schweier, G. Becker, New Holland Forage Harvester's Productivity in Short Rotation Coppice: Evaluation of Field Studies from a German Perspective, Int. J. For. Eng. 23 (2012) 82–88. doi:10.1080/14942119.2012.10739964.

[15] R. García Fernández, C. Pizarro García, A. Gutiérrez Lavín, J.L. Bueno de las Heras, J.J. Pis, Influence of physical properties of solid biomass fuels on the design and cost of storage installations, Waste Manag. 33 (2013) 1151–1157. doi:10.1016/j.wasman.2013.01.033.

[16] J. M. Ebeling, B. M. Jenkins, Physical and Chemical Properties of Biomass Fuels, Trans. ASAE. 28 (1985) 898–902. doi:10.13031/2013.32359.
[17] A. Sultana, A. Kumar, Optimal configuration and combination of multiple lignocellulosic biomass feedstocks delivery to a biorefinery, Bioresour. Technol. 102 (2011) 9947–9956. doi:10.1016/j.biortech.2011.07.119.
[18] M. Ebadian, M.E. Shedden, E. Webb, S. Sokhansanj, M. Eisenbies, T. Volk, J. Heavey, K. Hallen, Impact of Parcel Size, Field Shape, Crop Yield, Storage Location, and Collection Equipment on the Performance of Single-Pass Cut-and-Chip Harvest System in Commercial Shrub Willow Fields, BioEnergy Res. 11 (2018) 364–381. doi:10.1007/s12155-018-9902-7.
[19] K.L. Kenney, W.A. Smith, G.L. Gresham, T.L. Westover, Understanding biomass feedstock variability, Biofuels. 4 (2013) 111–127. doi:10.4155/bfs.12.83.
[20] C. Dupont, S. Rouge, A. Berthelot, D.D.S. Perez, A. Graffin, F. Labalette, C. Laboubee, J.-C. Mithouard, S. Pitocchi, Bioenergy II: Suitability of Wood Chips and Various Biomass Types for Use in Plant of BtL Production by Gasification, Int. J. Chem. React. Eng. 8 (2010) A74.
[21] M. Ebadian, S. Sokhansanj, E. Webb, Estimating the required logistical resources to support the development of a sustainable corn stover bioeconomy in the USA: Modeling and analysis: logistical resource assessment for the corn stover bioeconomy in the USA, Biofuels Bioprod. Biorefining. 11 (2017) 129–149. doi:10.1002/bbb.1736.
[22] T. Böhm, H. Hartmann, Bulk Density Determination of Solid Biofuels, Landtechnik. 60 (2005) 158–159.
[23] E. Valter Francescato, L.Z. Bergomi, C. Metschina, C. Schnedl, N. Krajnc, P. Kajetan Koscik, G. Nocentini, S. Stranieri, Wood Fuels Handbook, EIE/07/054. Italian Agriforestry Energy Association, Legnaro, Italy, 2008.
[24] E. Monedero, J. Hernández, R. Collado, Combustion-Related Properties of Poplar, Willow and Black Locust to be used as Fuels in Power Plants, Energies. 10 (2017) 997. doi:10.3390/en10070997.
[25] R. Harris, D. Phillips, Density of selected wood fuels, Research Paper 61 (Internet), Research Division Georgia Forestry Commission, Dry Branch, GA, USA, 1986. (cited 2018 Oct 15). Available from: http://www.gfc.state.ga.us/resources/publications/research-papers/GFRP61.pdf.
[26] C. Bush, T. Volk, M. Eisenbies, Planting rates and delays during the establishment of willow biomass crops, Biomass Bioenergy. 83 (2015) 290-296. http://dx.doi.org/10.1016/j.biombioe.2015.10.008
[27] M. Eisenbies, O. Therasme, T. Volk, K. Hallen, Bulk Density Data of Willow Chips for Collection Vehicles Operating in a Short Rotation Willow Crop, SUNY-ESF, Syracuse, NY 13210, 2019, (cited 2019 Mar 12). Available from: https://doi.org/10.7910/DVN/MNRDW8.
[28] ASABE, Moisture measurement - ANSI/ASABE Standards S358.2, American Society of Agricultural and Biological Engineers, St Joseph, MI, 2012.
[29] M.H. Eisenbies, T.A. Volk, J. Posselius, S. Shi, A. Patel, Quality and Variability of Commercial-Scale Short Rotation Willow Biomass Harvested Using a Single-Pass Cut-and-Chip Forage Harvester, BioEnergy Res. (2014). doi:10.1007/s12155-014-9540-7.
[30] C. Mallows, Some comments on Cp, Technometrics. 15 (1973) 661–675.

[31] D.C. Montogomery, E.A. Peck, G.G. Vining, Introduction to linear regression analysis, 3td ed., Wiley Interscience, 2001.
[32] P. Jensen, H. Hartmann, T. Bohm, M. Temmerman, F. Rabier, M. Morsing, Moisture content determination in solid biofuels by dielectric and NIR reflection methods, Biomass Bioenergy. 30 (2006) 935–943. doi:10.1016/j.biombioe.2006.06.005.
[33] V. Ganesan, K.A. Rosentrater, K. Muthukumarappan, Flowability and handling characteristics of bulk solids and powders – a review with implications for DDGS, Biosyst. Eng. 101 (2008) 425–435. doi:10.1016/j.biosystemseng.2008.09.008.
[34] J.M. Craven, J. Swithenbank, V.N. Sharifi, Investigation into the Flow Properties of Coarse Solid Fuels for Use in Industrial Feed Systems, J. Powder Technol. 2015 (2015) 1–12. doi:10.1155/2015/786063.
[35] US DOT, Compilation of existing state truck size and weight laws, (Internet), US Department of Transportation, Washington DC, 2015. (cited 2018 Oct 15). Available from: https://ops.fhwa.dot.gov/freight/policy/rpt_congress/truck_sw_laws/truck_sw_laws.pdf.
[36] E.M. Searcy, R.J. Hess, Uniform-format feedstock supply system: A commodity-scale design to produce an infrastructure-compatible biocrude from lignocellulosic biomass, INL/EXT-10-20372 (Internet). Idaho National Laboratory, Idaho Falls, ID, 2010 (cited 2019 Mar 1) Available from: https://bioenergy.inl.gov/Reports/Uniform Format Feedstock Supply System.pdf.
[37] E. Searcy, P. Flynn, E. Ghafoori, A. Kumar, The relative cost of biomass energy transport, Appl. Biochem. Biotechnol. 137 (2007) 639-652.
[38] C. Whittaker, N.E. Yates, S.J. Powers, T. Misselbrook, I. Shield, Dry matter losses and quality changes during short rotation coppice willow storage in chip or rod form, Biomass Bioenergy. 112 (2018) 29–36. doi:10.1016/j.biombioe.2018.02.005.
[39] A. Banski, L. Dzurenda, Density and bulk density of green wood chips from dendromass of short rotation coppice grown on plantations, Acta Fac. Xylologiae. 56 (2014) 17–26.
[40] C. Ryu, Y.B. Yang, A. Khor, N.E. Yates, V.N. Sharifi, J. Swithenbank, Effect of fuel properties on biomass combustion: Part I. Experiments—fuel type, equivalence ratio and particle size, Fuel. 85 (2006) 1039–1046. doi:10.1016/j.fuel.2005.09.019.
[41] A. Gendek, M. Aniszewska, K. Chwedoruk, Bulk density of forest energy chips, Ann Wars. Univ Life Sci. 67 (2016) 101–111.

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Figure Captions

Figure 1. Relationship between as received and dry bulk densities for willow biomass chips as determined using the ISO 17828:2015 method. Samples were collected from three dump wagons on a commercial-scale harvesting operation: small (≈10 m^3), fixed (≈30 m^3) and walking (≈30 m^3) dump wagon (as described in the methods).

Figure 2. Relationship between the as received bulk density (ISO 17828:2015) standing biomass, willow height, and moisture content. Symbols for individual points are shaded relative to increasing height (top), or increasing moisture content (bottom).

Figure 3. Relationship between the as received bulk densities measured directly for the dump bodies of three collection vehicles to standard bulk densities (ISO 17828:2015) and effective bulk density (ratio of weight to the fixed dump body volume). Samples were collected from three dump wagons on a commercial-scale harvesting operation: small (≈10 m^3), fixed (≈30 m^3) and walking (≈30 m^3) dump wagon (as described in the methods).

Figure 4. Relationship between the as received bulk densities measured directly for the dump bodies of three collection vehicles and the weight of individual loads. Samples were collected from three dump wagons on a commercial-scale harvesting operation: small (≈10 m^3), fixed (≈30 m^3) and walking (≈30 m^3) dump wagon (as described in the methods). Lines are indicative of the significant influence of standing biomass on actual bulk density for the walking and fixed dump wagons based on a multiple linear regression model (Model 4).

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Table 1. Bulk density values from found literature on shrub willow, and selected literature for other SRWC and generic wood chips.

Material	Context	Source or Method	Reporting Method	Sample Size	BD_d	Reported Moisture Content	BD_{ar}[a]	Reference
WILLOW								
Fresh Chips	Pile Study	Undescribed	Dry	0.0026 m3	173	56	393	[38]
Stored Chips	Pile Study	Undescribed	Dry	0.0026 m3	144	44	257	[38]
Willow Chips	Willow Varieties	Undescribed	Dry	Undescribed	138-184	44-51	250-365	[9]
Willow Chips	Feedstock Quality	EN 15103	As received	Undescribed	171	52	356	[39]
Ground Willow	Feedstock Quality	EN 15103	Dry	"bottle"	215[b]	52	447**	[24]
Willow Chips	Combustion	Undescribed	Dry	Undescribed	181[c]	7.2	195	[40]
OTHER SRWC								
Eucalyptus Chips	Feedstock Quality	EN 15103	Undescribed	50 g	200+-28	11	224+-31	[15]
Poplar Chips	Feedstock Quality	EN 15103	As received	Undescribed	144	52	301	[39]
Poplar Chips	Feedstock Quality	EN 15103	Undescribed	Undescribed	192	53		[24]
SRWC Chips	Moisture Determination	Undescribed	Dry	0.05 m^3	130-157	30	186-224	[32]
GENERIC WOOD CHIPS								
"Wood" chips	Feedstock Quality	EN 15103	Undescribed		189+-23	26	255+-31	[15]
Hardwood whole tree chips	Feedstock Quality	Described	As received	16 m^3	207-218	35	318-335	[25]
Hog fuel chips	Feedstock Quality	Described	As received	16 m^3	181-236	12	206-269	[25]
Wood chips	Economics	Cited	Undescribed	Undescribed	180-340	50		[17]
Wood chips	Economics	Cited	Undescribed	Undescribed	168	30		[17]
Forest Chips (birch, pine, spruce)	Equipment	EN 15103	Dry and As Received	0.05 m^3	140-246	30	202-261	[41]

[a] Where BD_{ar} is not reported, it is calculated using the provided moisture Content. $BD_d/(1\text{-}MC\%/100)$

[b] Exceptionally small particle sizes 98% of sample less than 3.15mm

[c] Low moisture as received

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Table 2. Particle size distribution of chips collected from 7 fields relative to the ISO 17225-4 standard for graded wood chips. Field name is an arbitrary designation that was based on distance to landing and are sorted in this table based on stand biomass.

Field Name	Height	Stand biomass			Fines Fraction	Coarse Fraction				Maximum Length	Maximum Cross Section
				Limit	≤3.15 mm	≤15.9mm	≤25.4mm	≤31.8mm	≤63mm	≤200mm	≤6cm^2
				Standard	≤10%				≥90%	100%	100%
	m	Mg ha^{-1}			%	%	%	%	%	mm	cm^2
	$\bar{x}$	$\bar{x}$	σ								
B	6.9	69.9	10.6		3	38	56	84	96	120	4
A	6.4	67.7	6.2		.	.	.	.	98	120	4
D	7.0	51.1	5.7		2	16	26	45	96	144	4
C	6.2	48.0	13.7		1	12	18	45	96	140	4
G	5.0	40.6	4.1		5	42	57	87	98	110	4
E	5.0	29.4	5.8		1	14	23	62	97	147	4
F	4.7	16.2	4.3		3	33	41	75	96	142	4

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Table 3. Mean bulk density values, determined three ways, for three dump wagons. Letters indicate significant differences within columns.

Wagon Type	Number of Loads	Wagon Volume	Mean Overfill	Mean Unfilled Void	Load Weight		Moisture Content		Bulk Density as received						Bulk Density (0% Moisture)					
									BD_{ar}^{ISO}		BD_{ar}^{act}		BD_{ar}^{eff}		BD_{ar}^{ISO}		BD_{ar}^{act}		BD_{ar}^{eff}	
	n	----------m^3----------			------kg-------		-----%------		----------kg m^3----------						----------kg m^3----------					
					$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ
Small	16	10.3	2.4	2.3	3260	312	53.0 A	3.4	258 A	28	303 A	32	317 A	30	136 A	19	159 A	22	169 A	24
Fixed Axle	19	28.0	4.6	2.4	6412	778	53.1 A	3.3	265 A	22	212 B	27	229 B	28	139 A	16	111 B	16	121 B	17
Walking Axle	18	29.5	4.1	2.9	6340	1465	52.5 A	2.5	264 A	25	216 B	30	215 B	50	138 A	16	113 B	16	113 B	26

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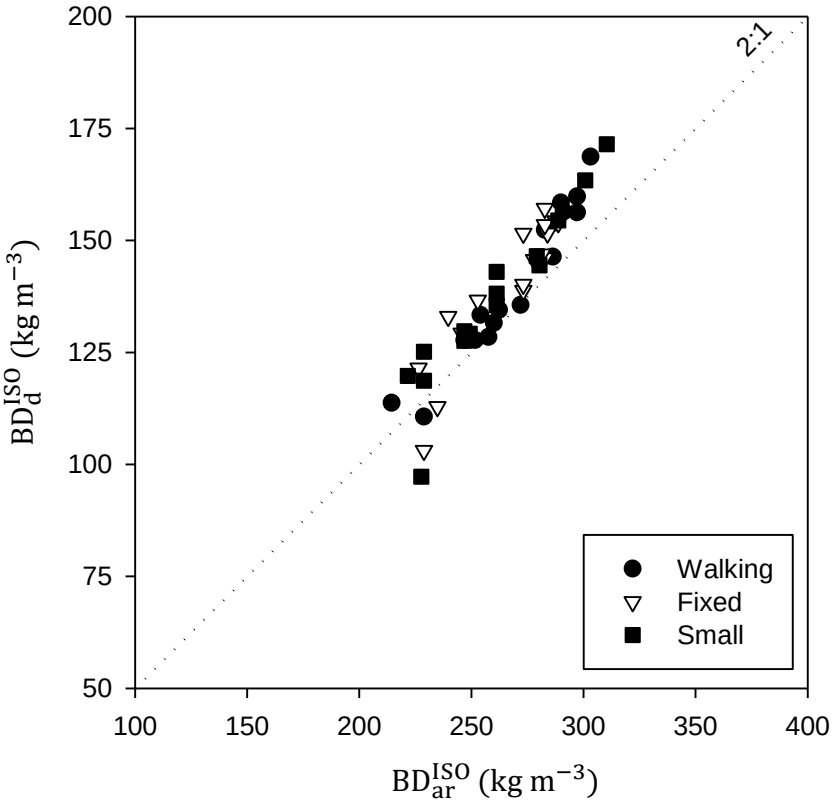


Figure 1. Relationship between as received and dry bulk densities for willow biomass chips as determined using the ISO 17828:2015 method. Samples were collected from three dump wagons on a commercial-scale harvesting operation: small (≈10 m^3), fixed (≈30 m^3) and walking (≈30 m^3) dump wagon (as described in the methods).

THIS IS A PREPRINT OF THE ABOVE REFERENCE

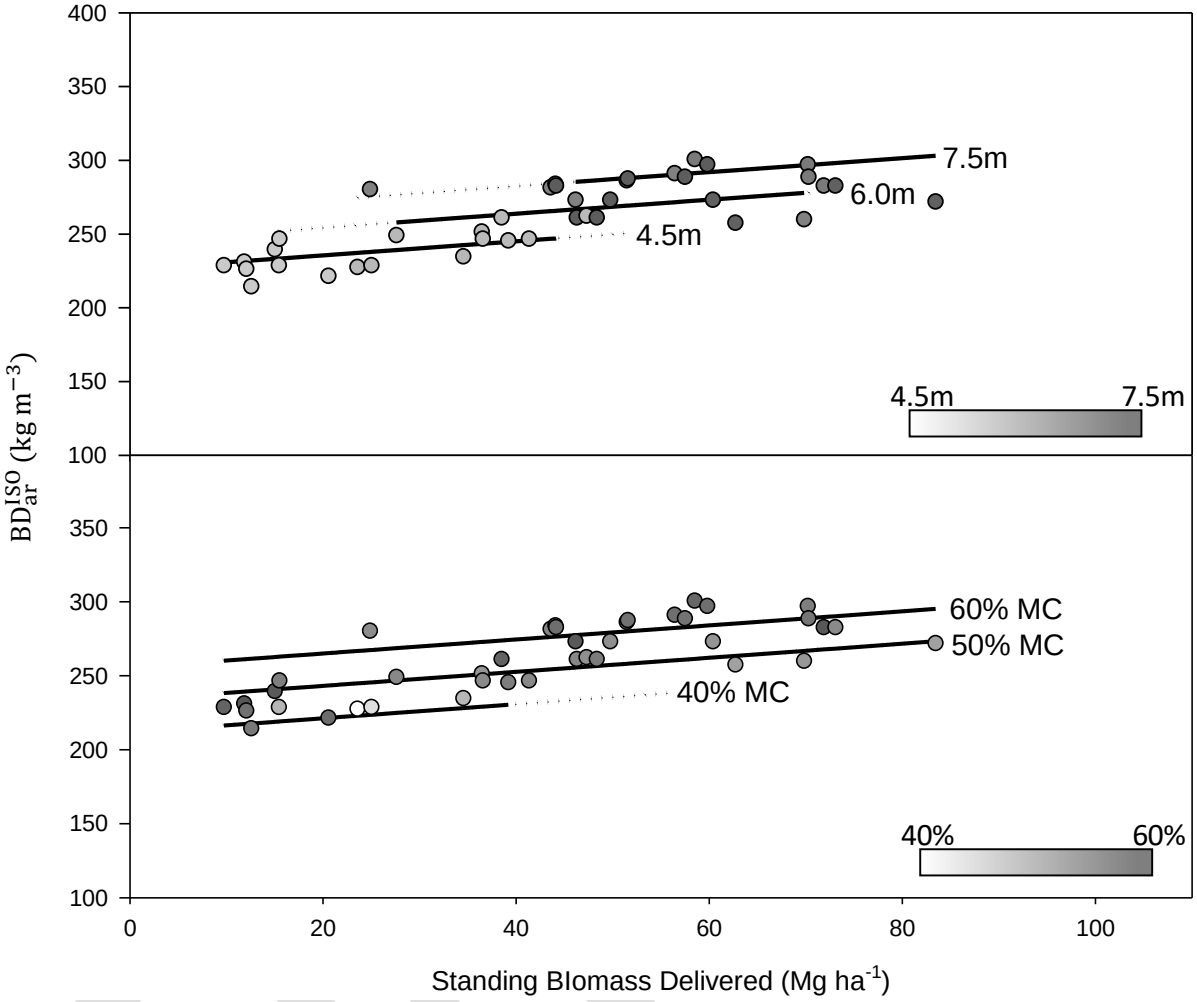


Figure 2. Relationship between the as received bulk density (ISO 17828:2015) standing biomass, willow height, and moisture content. Symbols for individual points are shaded relative to increasing height (top), or increasing moisture content (bottom).

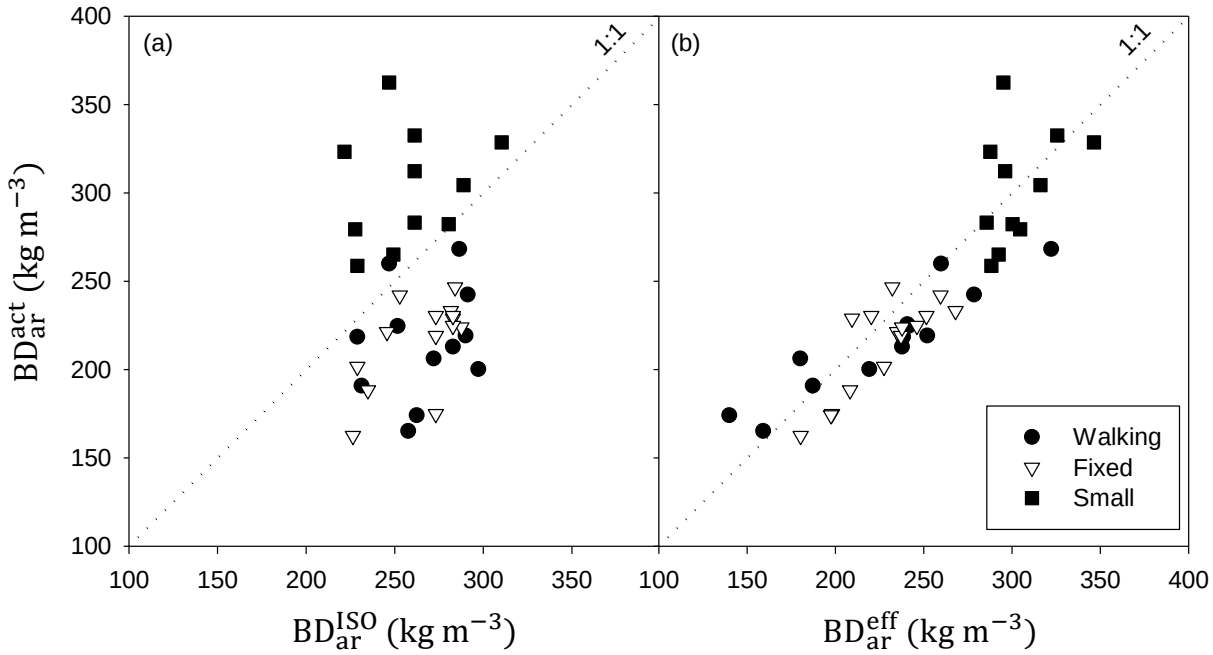


Figure 3. Relationship between the as received bulk densities measured directly for the dump bodies of three collection vehicles to standard bulk densities (ISO 17828:2015) and effective bulk density (ratio of weight to the fixed dump body volume). Samples were collected from three dump wagons on a commercial-scale harvesting operation: small (≈10 m^3), fixed (≈30 m^3) and walking (≈30 m^3) dump wagon (as described in the methods).

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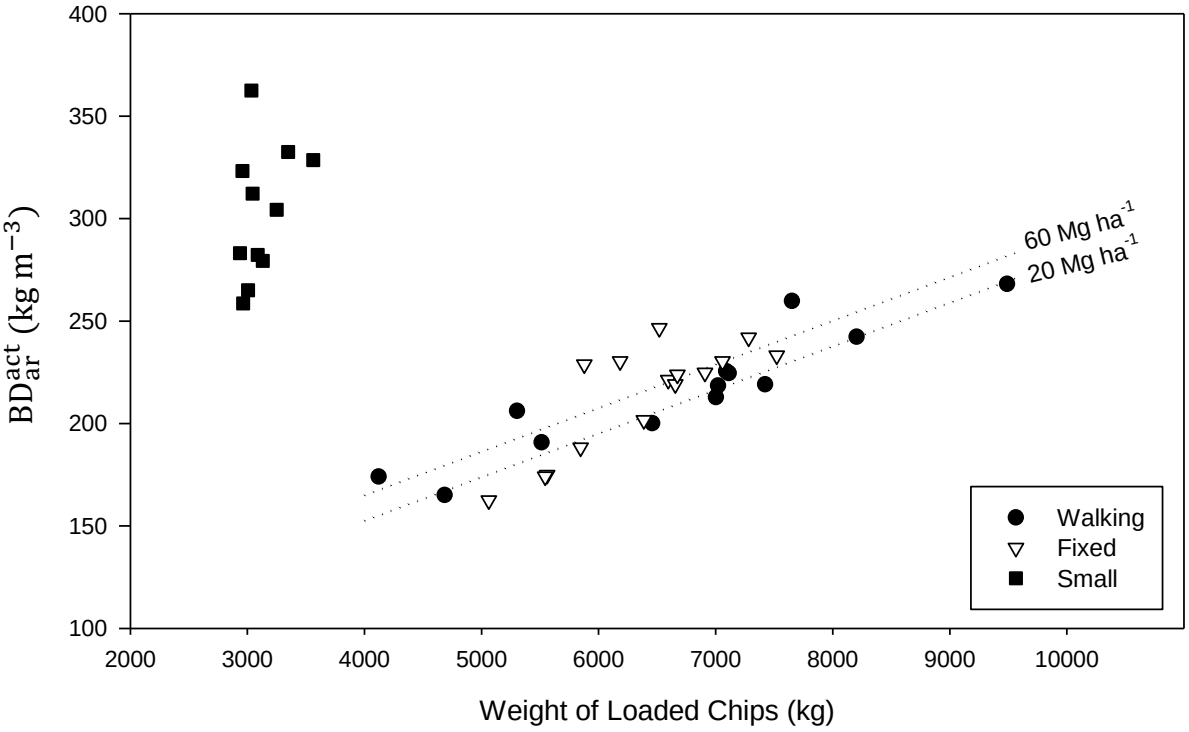


Figure 4. Relationship between the as received bulk densities measured directly for the dump bodies of three collection vehicles and the weight of individual loads. Samples were collected from three dump wagons on a commercial-scale harvesting operation: small (≈10 m^3), fixed (≈30 m^3) and walking (≈30 m^3) dump wagon (as described in the methods). Lines are indicative of the significant influence of standing biomass on actual bulk density for the walking and fixed dump wagons based on a multiple linear regression model (Model 4).

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